

Ishani Kathuria

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AI/ML engineer with 2 years building production LLM systems at AWS, now completing an **MS in Applied Artificial Intelligence (expected May 2027)** with research in RAG and LLM evaluation. 4 peer-reviewed publications. Available full-time from May 2027.

EXPERIENCE

Purdue University Northwest

Sep 2025 – Present

Artificial Intelligence Research Assistant

Hammond, Indiana, USA

- Conducting applied research on **Retrieval-Augmented Generation (RAG)** systems, focusing on adversarial defenses, retrieval quality, and hallucination reduction for LLM-based applications.

Amazon Web Services

Jul 2023 – Jul 2025

Software Development Engineer

Bangalore, Karnataka, India

- Built and deployed **LLM-based log summarization** for production services, cutting mean root-cause analysis time from 2.5 hours to 30 minutes.
- Designed internal **AI chatbots** for developer support workflows, improving resolution efficiency on routine operational queries by 40% for 200+ engineers.
- Automated region-build and deployment pipelines for 15+ OpenSearch services using AWS infrastructure tooling, reducing manual intervention by 80% and accelerating launch timelines by 3 weeks.
- Developed proactive anomaly and bug detection systems to catch failures pre-release, reducing externally reported customer issues by 30%.

Earlier Experience

2021 – 2022

- Artificial Intelligence Automation Intern, Deloitte** (2022) — Built hybrid RPA solutions with Automation Anywhere and Python to automate complex internal reporting tasks.
- Data and Algorithm Lead, Planet E-Com Solutions** (2022) — Engineered Python web-scraping and data-processing pipelines that improved the accuracy of education consultation software.
- AI Developer & Lead, Docplus Online** (2021) — Built Python AI pipelines for translation and gender-classification models, coordinating technical implementation across 40+ developers.

PROJECTS

TrustworthyRAG

Python, LangChain, knowledge graphs

- Designed a query-adaptive learned fusion (QALF) mechanism that dynamically routes queries across vector, graph, and keyword retrieval systems in a multimodal knowledge graph that has unstructured text and images to enable multi-hop reasoning over multiple documents.

AutoRedTeam

Python, React, multi-provider LLM APIs

- Engineered a multi-agent adversarial evaluation framework (Attacker, Target, Judge) to stress-test LLM safety and robustness with provider-agnostic LLM integrations (GPT-4, Gemini, Llama 3) with a real-time React visualization interface.

DeepFakeGuard

Transformers.js, TypeScript

- Built a client-side AI forensic tool that detects synthetic audio entirely in-browser with Transformers.js, removing server round-trips and preserving user privacy and optimized streaming inference pipelines to sub-second latency with no server-side dependencies.

EDUCATION

Purdue University Northwest

Aug 2025 – May 2027 (expected)

Master of Science in Applied Artificial Intelligence (GPA: 4.0/4.0)

Hammond, Indiana, USA

Amity University

Jul 2019 – Jul 2023

Bachelor of Technology in Artificial Intelligence (GPA: 9.09/10.0)

Noida, Uttar Pradesh, India

SKILLS & CERTIFICATIONS

- Generative AI:** RAG, fine-tuning, multi-agent systems, prompt engineering, LLM evaluation, LangChain, Hugging Face, Ollama; GPT-4, Claude, Gemini, Gemma, Llama 3, Mistral, Amazon Nova, FAISS/pgvector, RAGAS/LangSmith, vLLM, FastAPI
- Programming:** Python, Go, TypeScript, Java, SQL, Bash
- ML, Data & Web:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Tableau, React, Flask, Django
- Cloud & MLOps:** AWS (Bedrock, Lambda, ECS, Step Functions, CloudFormation, CloudWatch), Azure, Docker, CI/CD
- Certifications:** AWS Certified AI Practitioner, AWS Certified Cloud Practitioner ([Credly](#))

PUBLICATIONS (4 PEER-REVIEWED)

- [Conversational AI for Supporting Children with Autism](#) (Springer, 2025)
- [Temperature Based Food Recommendation using AI](#) (IEEE, 2023)
- [Deep Learning in Healthcare](#) (Springer, 2023)
- [PM2.5 Prediction using Azure ML Studio](#) (IEEE, 2022)